

Digital Forensics Report

1 Introduction

The purpose of this report is to provide corroborative evidence to illegal hacking practices including but not limited to hacking software, evidence of their use, and any data that might have been generated, and tie the computer to the suspect, Greg Schardt. The facts presented in this report are those within the preparer's own area of expertise and knowledge and do not extend to matters and knowledge outside of such expertise.

2 Forensic Examination

2.1 Tools

AUTOPSY

Autopsy was the software used for analyzing digital data found in a disk image generated from the Dell CPi Notebook Computer, serial # VLQLW. Autopsy is an open source digital forensics software which provides investigators with the tools to conduct complex investigations with accuracy and efficiency. Autopsy provides the ability to manage large volumes of computer evidence and view computer drive contents including files, operating system artifacts, file system artifacts, and deleted files or file fragments located in file slack or unallocated space. To the expert's knowledge, there is no software in use above that possesses any material issues whose severity would invalidate the findings thereof.

2.2 Evidence Classes

Evidence Class	Description
1	Hacking software
2	Evidence of use of hacking software
3	Data generated by use of hacking software
4	Connection between computer and suspect

2.2.1 Evidence Class 1 - Hacking Software

The following hacking software was found:

1. NetCat

Netcat is a featured networking utility which reads and writes data across network connections, using the TCP/IP protocol.

It is designed to be a reliable "back-end" tool that can be used directly or easily driven by other programs and scripts. At the same time, it is a feature-rich network debugging and

exploration tool, since it can create almost any kind of connection you would need and has several interesting built-in capabilities

Exhibit A - screenshot of NetCat program files installed in Schardt's computer.

▲ Name	S	C	O	Modified Time	Change Time	Access Time	Created Time
 nc110.gz			1	2000-06-26 12:59:18 PDT	2004-08-20 08:18:41 PDT	2004-08-20 08:18:41 PDT	2004-08-20 08:18:41 PDT
 nc110.tgz			1	1998-11-16 11:03:30 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
 nc110.tgz			1	1999-01-03 07:58:32 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
 nc110.tgz			1	1998-11-16 11:03:30 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
 nc110.tgz			1	1999-01-03 07:58:32 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
 netbase.tgz			1	1999-01-03 08:32:36 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
 netbase.tgz			1	1999-01-03 09:32:35 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
 netbase.tgz			1	1999-01-03 08:32:36 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
 netbase.tgz			1	1999-01-03 09:32:35 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
 netcat11nt.zip			0	2000-06-13 13:05:10 PDT	2004-08-20 08:19:41 PDT	2004-08-20 08:19:41 PDT	2004-08-20 08:19:41 PDT

2. John The Ripper

John The Ripper is a password cracker, currently available for UNIX, DOS, WinNT / Win95. Its primary purpose is to detect weak UNIX passwords.

Exhibit B - screenshot of John The Ripper program files installed in Schardt's computer.

▲ Name	S	C	O	Modified Time	Change Time	Access Time	Created Time
 john-1.6.tar	▼		1	1998-12-02 16:29:50 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
 john-1.6.tar.gz	▼		1	2000-10-21 07:41:10 PDT	2004-08-20 08:19:09 PDT	2004-08-20 08:19:09 PDT	2004-08-20 08:19:09 PDT
 john-16w.zip	▼		1	2000-03-24 12:29:04 PST	2004-08-20 08:19:47 PDT	2004-08-20 08:19:47 PDT	2004-08-20 08:19:46 PDT
 john-1~1.gz	▼		1	1999-01-07 11:56:30 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
 john-1~1.gz	▼		1	1999-01-07 11:56:30 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
 john-k6.zip	▼		1	1998-12-03 03:19:08 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
 john-k6.zip	▼		1	1998-12-03 03:19:08 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
 john-mmx.zip	▼		1	1998-12-03 03:19:08 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
 john-mmx.zip	▼		1	1998-12-03 03:19:08 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00

3. Brutus

Brutus is an online or remote password cracker. It is a remote interactive authentication agent. Brutus is used to recover valid access tokens (usually a username and password) for a given target system. Examples of a supported target system might be an FTP server, a password protected web page, a router console a POP3 server, etc.

Exhibit C - screenshot of Brutus program files installed in Schardt's computer.

▲ Name	S	C	O	Modified Time	Change Time	Access Time	Created Time
 brutus.zip			1	2000-10-19 13:41:40 PDT	2004-08-20 08:19:13 PDT	2004-08-20 08:19:13 PDT	2004-08-20 08:19:12 PDT

4. Cain & Abel

A windows-only password recovery handles an enormous variety of tasks. It can recover passwords by sniffing the network, cracking encrypted passwords.

5. **Ethereal**
A packet sniffer, an application that can capture all packets sent through a network to then interpret them and see what tasks are being carried out on the network.
6. **123 Write All Stored Passwords**
A program that reads the Windows .PWL file and tells you all the stored Windows user passwords.
7. **Anonymizer**
A tool that masks your IP address and location, making it harder for websites to track your activities and identify you. It works by routing your internet traffic through a proxy server, which then forwards your requests to the destination website anonymously.
8. **Look@LAN**
A network discovery tool.
9. **NetStumbler - wireless access point discovery tool**
Netstumbler.com offers freeware written by Marius Miner that allows anyone with an 802.11b wireless LAN access card to detect and analyze wireless LANs. The software is known for its ability to identify open wireless access points, known as "wardriving," and is useful for verifying network setups, finding coverage issues, and detecting interference. Although it is not open-source, Netstumbler is free to download and requires Windows 2000, XP, or newer operating systems.
10. **CuteFTP - FTP software**

Exhibit D - screenshot of installed programs in Schardt's computer including: Cain & Abel, Ethereal, 123 Write All Store Passwords, Anonymizer, Look@LAN and NetStumbler, CuteFTP.

Installed Programs						
Table		Thumbnail	Summary			
Source Name	S	C	O	Program Name	Date/Time	Data Source
software			0	123 Write All Stored Passwords	2004-08-20 15:13:08 PDT	SCHARDT.000
software			0	AddressBook	2004-08-19 22:31:51 PDT	SCHARDT.000
software			0	Anonymizer Bar 2.0 (remove only)	2004-08-20 15:05:09 PDT	SCHARDT.000
software			0	Branding	2004-08-19 22:37:31 PDT	SCHARDT.000
software			0	Cain & Abel v2.5 beta45	2004-08-20 15:05:58 PDT	SCHARDT.000
software			0	Connection Manager	2004-08-19 22:21:41 PDT	SCHARDT.000
software			0	CuteFTP	2004-08-20 15:09:02 PDT	SCHARDT.000
software			0	CuteHTML	2004-08-20 15:09:03 PDT	SCHARDT.000
software			0	DirectAnimation	2004-08-19 22:31:52 PDT	SCHARDT.000
software			0	DirectDrawEx	2004-08-19 22:31:32 PDT	SCHARDT.000
software			0	Ethereal 0.10.6 v.0.10.6	2004-08-27 15:29:19 PDT	SCHARDT.000
software			0	Faber Toys v.2.4 Build 216	2004-08-20 15:07:25 PDT	SCHARDT.000
software			0	Fontcore	2004-08-19 22:31:32 PDT	SCHARDT.000
software			0	Forté Agent	2004-08-20 15:08:19 PDT	SCHARDT.000
software			0	ICW	2004-08-19 22:31:51 PDT	SCHARDT.000
software			0	IE40	2004-08-19 22:31:32 PDT	SCHARDT.000
software			0	IE4Data	2004-08-19 22:31:32 PDT	SCHARDT.000
software			0	IE5BAKEX	2004-08-19 22:31:32 PDT	SCHARDT.000
software			0	IEData	2004-08-19 22:31:32 PDT	SCHARDT.000
software			0	Look@LAN 2.50 Build 29	2004-08-25 15:56:11 PDT	SCHARDT.000
software			0	MPlayer2	2004-08-19 23:04:36 PDT	SCHARDT.000
software			0	Microsoft NetShow Player 2.0	2004-08-19 23:04:36 PDT	SCHARDT.000
software			0	MobileOptionPack	2004-08-19 22:31:32 PDT	SCHARDT.000
software			0	NetMeeting	2004-08-19 22:31:52 PDT	SCHARDT.000
software			0	Network Stumbler 0.4.0 (remove only)	2004-08-27 15:12:15 PDT	SCHARDT.000

2.2.2 Evidence Class 2 - Evidence of use of hacking software

The following screenshots display the use of the following hacking software found in Mr. Schardt's computer including Brutus, John The Ripper and NetCat.

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time
BINDERX.EXE	✓		1	1995-11-03 12:26:10 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
BINDIN.EXE			0	1990-07-16 10:30:28 PDT	2004-08-20 08:21:06 PDT	2004-08-20 08:21:06 PDT	2004-08-20 08:21:06 PDT
BrutusA2.exe	✓		1	2000-01-29 12:13:34 PST	2004-08-20 08:19:48 PDT	2004-08-20 08:19:48 PDT	2004-08-20 08:19:48 PDT
BrutusA2.exe	✓		1	2000-01-28 16:13:34 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
BrutusA2.exe	✓		1	2000-01-28 16:13:34 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time
isignup.exe			1	2001-08-23 11:00:00 PDT	2004-08-19 15:44:23 PDT	2004-08-19 15:44:23 PDT	2004-08-19 15:29:08 PDT
iun6002.exe			1	2004-08-25 08:55:27 PDT	2004-08-26 07:58:14 PDT	2004-08-25 08:56:09 PDT	2004-08-25 08:56:09 PDT
john.exe			1	1998-12-03 03:19:08 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
john.exe			1	1998-12-03 03:14:06 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
john.exe			1	1998-12-03 03:14:06 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
john.exe			1	1998-12-03 03:19:08 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
john.exe			1	1998-12-03 03:14:06 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
john.exe			1	1998-12-03 03:14:06 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time
nbtstat.exe			1	2001-08-23 11:00:00 PDT	2004-08-19 10:02:17 PDT	2004-08-19 15:45:38 PDT	2001-08-23 11:00:00 PDT
nc.exe			1	1998-01-04 09:37:34 PST	2004-08-20 08:18:14 PDT	2004-08-20 08:18:14 PDT	2004-08-20 08:18:14 PDT
nc.exe			1	1998-01-03 13:37:34 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time
NTUserlist.txt			0	2001-01-06 13:14:14 PST	2004-08-20 08:18:15 PDT	2004-08-20 08:18:15 PDT	2004-08-20 08:18:15 PDT
NTUserlist.txt			0	2001-01-04 10:43:46 PST	2004-08-20 08:19:18 PDT	2004-08-20 08:19:18 PDT	2004-08-20 08:19:18 PDT
OEWABlog.txt			0	2004-08-19 16:04:37 PDT	2004-08-19 16:04:37 PDT	2004-08-19 16:04:37 PDT	2004-08-19 15:37:49 PDT
ORDERFRM.TXT			1	2000-10-11 16:27:04 PDT	2004-08-20 08:18:10 PDT	2004-08-20 08:18:10 PDT	2004-08-20 08:18:10 PDT
PASSLIST.TXT			1	1997-02-17 13:44:24 PST	2004-08-20 08:18:15 PDT	2004-08-20 08:18:15 PDT	2004-08-20 08:18:15 PDT
PASSLIST.TXT			1	1997-02-16 16:44:24 PST	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
PUBKEY.TXT			0	2000-10-11 16:27:04 PDT	2004-08-20 08:18:10 PDT	2004-08-20 08:18:10 PDT	2004-08-20 08:18:10 PDT
READ.TXT			0	1998-05-14 17:47:30 PDT	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00
README.TXT			1	2000-10-11 16:27:04 PDT	2004-08-20 08:18:10 PDT	2004-08-20 08:18:10 PDT	2004-08-20 08:18:10 PDT

Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences

Strings
Extracted Text
Translation

Page: 1 of 1 Page
Matches on page: - of - Match
100%
Reset

ADMINISTRATOR
GUEST
ROOT
ADMIN
PASSWORD
TEMP
SHARE
WRITE
FULL
BOTH
READ
FILES
DEMO
TEST
ACCESS
USER
BACKUP
SYSTEM
SERVER
LOCAL

2.2.3 Evidence Class 3 - Data generated by use of hacking software

Under Mr. Evil's documents, the following files were found containing dictionary lists of passwords that may have been used to perform Brute Force cyber attacks when trying to crack target systems' end user passwords.

1. pocket-dic.gz

Name	S	C	O	Modified Time	Change Time	Access Time
pocket-dic	✓		0	1994-11-22 16:32:43 PST	0000-00-00 00:00:00	0000-00-00 00:00:00

Hex	Text	Application	File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations	Other
Strings Extracted Text Translation									
Page: 1 of 6 Page Matches on page: - of - Match 100% Reset									
#####									
# pockt dictionary world.std.com:/obi/									
#####									
a									
aardvark									
aback									
abacus									
abaft									
abalone									
abandon									
abandoned									
abase									
abash									
abate									
abatement									

2. Unix.dict.gz

Name	S	C	O	Modified Time	Change Time
 Unix.dict			1	1992-02-16 22:16:26 PST	0000-00-00 00:00:00

Hex

Text

Application

File Metadata

OS Account

Data Artifacts

Analysis Results

Strings

Extracted Text

Translation

Page: 1 of 7 Page

←

→

 Matches on page: - of - Match

←

→

 100%

🔍

🔍

10th

1st

2nd

3rd

4th

5th

6th

7th

8th

9th

a

a&m

a&p

a's

aaa

3. Words-english.gz

2.2.4 Evidence Class 4 - Connection between computer and suspect

The following screenshot shows the Microsoft Windows XP license owner as Greg Schardt. This piece of evidence connects Mr. Greg Schardt to the Dell CPi notebook computer.

Exhibit - Screenshot of Mr. Shcardt's computer license information.

Source Name	S	C	O	Name	Program Name	Processor Architecture	Temporary Files Directory	Path	Product ID	Owner
img_SCHARDT.000				N-1A9ODN6ZXK4LQ	Microsoft Windows XP	x86	%SystemRoot%\TEMP	C:\WINDOWS	55274-640-0147306-23684	Greg Schardt
No comments found										
Hex Text Application Source File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences										
Result: 1 of 1 Result < >										
Type	Value									Source(s)
Name	N-1A9ODN6ZXK4LQ									Recent Activity
Program Name	Microsoft Windows XP									Recent Activity
Processor Architecture	x86									Recent Activity
Temporary Files Directory	%SystemRoot%\TEMP									Recent Activity
Path	C:\WINDOWS									Recent Activity
Product ID	55274-640-0147306-23684									Recent Activity
Owner	Greg Schardt									Recent Activity
Organization	N/A									Recent Activity
Source File Path	/img_SCHARDT.000									
Artifact ID	-9223372036854775711									

2.3 ANALYSIS & CONCLUSION

The above evidence found in Mr. Schardt's computer demonstrates the presence and use of hacking software pertaining to intercepting internet traffic. Based on our findings pertaining to the computer's Operating System license, we can attest that the computer that was confiscated (Dell CPi notebook computer, serial # VLQLW) was indeed Mr. Schardt's. Although no stolen personal information was found, the presence of CuteFTP indicates that there could be other files relevant to the investigation if such software could be accessed.

In conclusion, based on our findings, we believe that Mr. Shchardt was breaking the law by illegally attempting to connect to weak Wi-Fi connections by the use of brute force attacks; however, no evidence was found pertaining to stealing credit card numbers or other personal information.